An Axiomatic Concurrency Model

45

S A +

S A

R A +

R

A

S B +

S B –

C A +

C A –

R B +

R B _

C B +

C

_

po

po

r

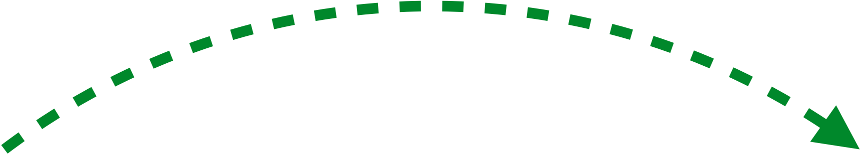
r

po

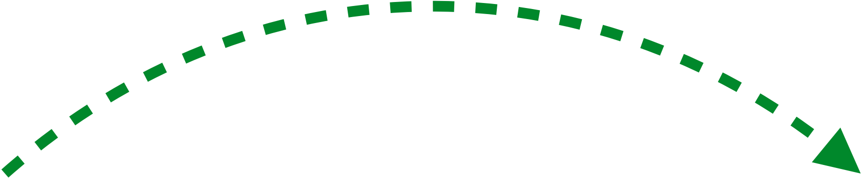
po

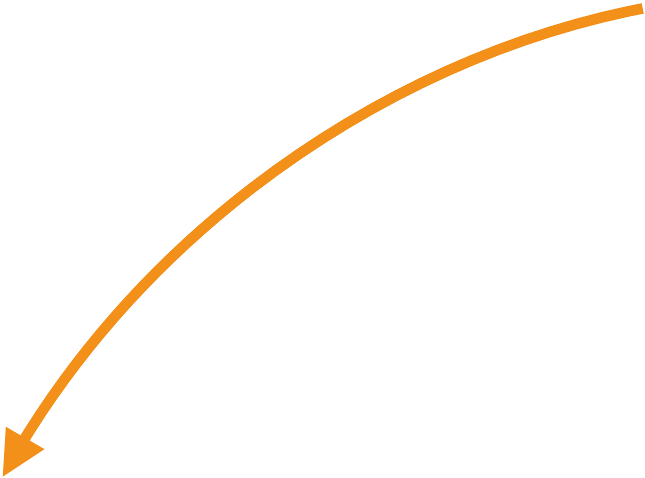
po

po

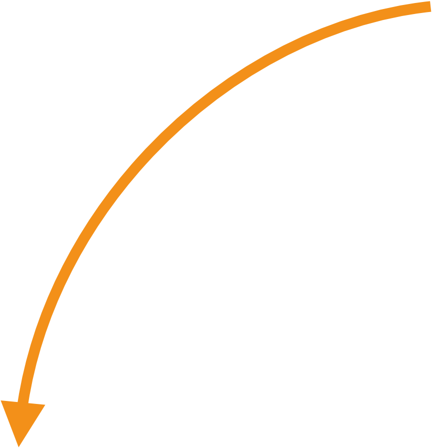


r

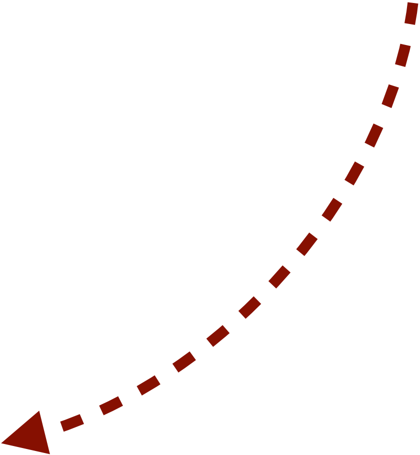


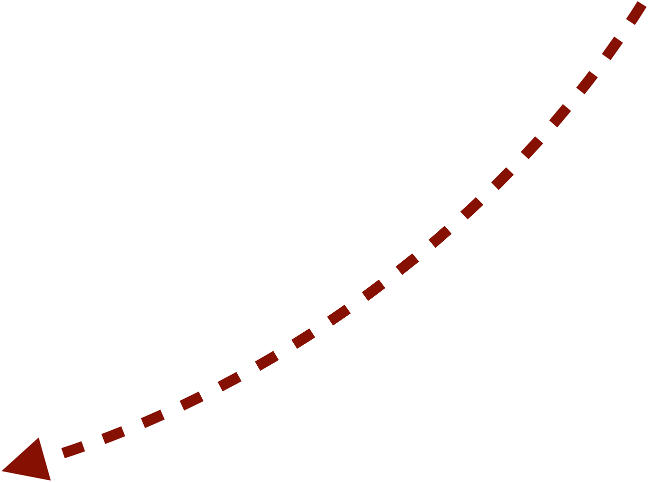


COsrc



COlog





hb

hb

```
update(src, +100)
```

```
update(src, -100)
```

r

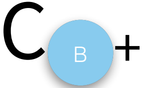
Valid executions are acyclic in the transitive closure of the union of relations

```
def update(account: cown[Account], amount: int) {
  when(account) { A
    if(account.balance + amount > 0) {
      account.balance += amount
      val id = account.id()
      when(log) { B log.record(id, amount) }
} } }
```

R B +

R

B

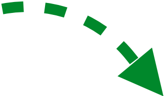

C
B
+

C

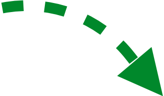
B

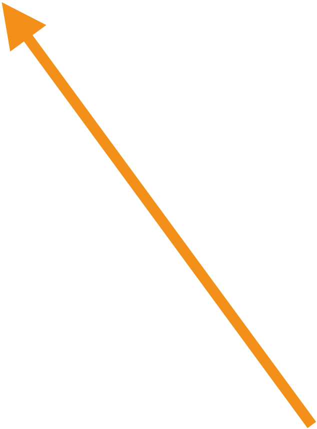
po

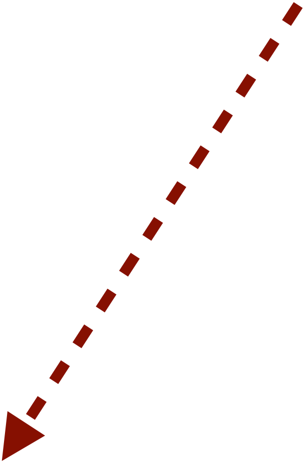
po



COlog







hb

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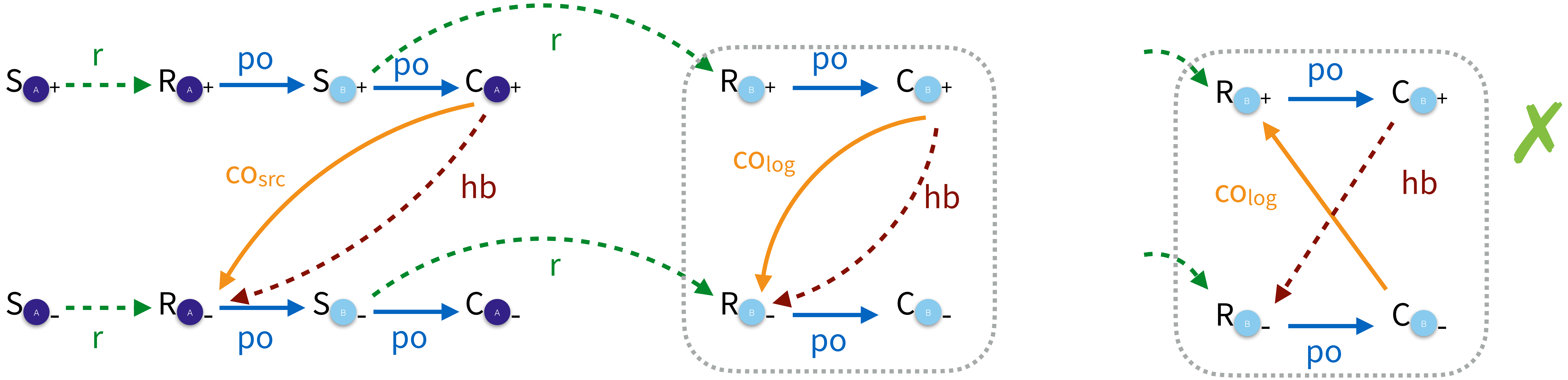

What does this get us?

- What does it mean to say all of the executions the semantics provide are even correct?